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1 RAPIDRADIO'S INDIGENOUS RFID READER PLATFORM THAT ENABLES NUMEROUS APPLICATIONS

RapidRadio's innovation is not another RFID reader but a complete electronics platform that enables manufacturers to develop their own solutions faster and with greater flexibility. The result is a homegrown technology now deployed in more than 500 libraries and numerous industrial applications.

Every time a FASTag lane opens automatically, or an EV charging station recognises a user, or a vehicle enters a gated community without stopping, an RFID reader works silently in the background. Although such systems are now part of everyday life, the electronics that power them are often imported from China, Europe, or the United States. Ahmedabad-based RapidRadio has spent nearly two decades reducing this dependence by developing indigenous RFID electronics platforms.

Founded by engineers Dhaval H. Kotecha, Nirav Karia, and Dushyant Upadhyay, RapidRadio focuses on developing core electronic architectures that enable manufacturers to build RFID-based products without relying on imported hardware. The company was established in 2006 within Nirma University's incubation ecosystem, when RFID technology was still relatively unfamiliar in India.

RapidRadio's key innovation is its integrated ultra-high frequency (UHF) RFID module. This module combines RFID reading capability, embedded processing, communication interfaces, memory management, and control functions into a single platform. Instead

of spending months integrating multiple electronic subsystems, manufacturers can use the module as a ready-made foundation and focus on their end applications. The technology supports smart parking systems, vehicle-access solutions, EV charging stations, industrial tracking platforms, and other RFID-enabled products while significantly reducing development complexity and time-to-market.

Developing such a platform was challenging. The founders first had to master antenna design. Although chip manufacturers supplied technical documentation for reader electronics,

practical expertise in antenna tuning was limited. The team's efforts resulted in a functioning RFID reader-and-antenna combination within about seven months, an achievement that often requires much larger organisations and longer timelines.

The next challenge emerged when the technology was exposed to real-world conditions. RFID systems frequently struggle in environments containing metal structures because metallic surfaces distort radio-frequency signals and reduce reading accuracy. RapidRadio adopted a different strategy by optimising reader-and-antenna combinations

directly in deployment environments rather than under ideal testing conditions. This enabled reliable performance in factories, warehouses, industrial facilities, and other challenging locations where conventional RFID systems often underperform.

These experiences led to RapidRadio's flagship product: a highly integrated UHF RFID module. RapidRadio's platform serves as a complete electronics building block. A key differentiator is its flexibility. Because the technology is designed and controlled locally, customers can request modifications to hardware form factors, firmware



RFID modules